ROADWATCH · OXHACK

Combined Run-of-Show

Both presenters · synchronised timeline, cues and fallback route

Event: OxHack · **Limit:** 5:00 · **Planned:** 4:30 + 0:30 contingency · **Story order:** real data → processing → prediction → verified evidence → actionable decision → limitations and next steps

Roles

- **Presenter 1 (P1)** speaks throughout, using `docs/final/presentation_speech.md`.
- **Presenter 2 (P2)** operates the laptop silently (slides and dashboard), using `docs/final/demo_instructions.md`.

This table is the master copy. Both documents use the same slide order, timings, cue phrases and figures.

Example week: the demo replays the saved forecast for **Monday 16 March 2026**. It is a retrospective example, not a live maintenance recommendation.

Setup checklist (before the clock starts)

#	Check	Owner
1	Dashboard running: `.venv\Scripts\python.exe -m streamlit run app.py --server.headless true --server.port 8501`; health check returns `ok`	P2
2	Edge full screen at `http://localhost:8501/?week=2026-03-16&view=map`; every demo step clicked once to warm the cache, then reloaded; map and worklist filling the screen	P2
3	Opening state confirmed: **16 Mar 2026 · 235 · 1% · 47 / 71 / 117**; inspection list = *Strict single-block LightGBM, top-K*; rank 1 = **17 AVE, ZIP 11204, 0.266**	P2
4	`RoadWatch_OxHack_Pitch.pptx` slideshow on Slide 1 (**F5**); Alt+Tab to the browser and back tested once	P2
5	Backup tab open with `RoadWatch_OxHack_Pitch.pdf`	P2
6	Optional: `scripts\final\verify_run_of_show.py` returns `problems: []`	P2
7	Stopwatch or phone timer ready; speech printed or on a second screen	P1

#	Check	Owner
8	Agreed hand signal: P2 raises a hand if the dashboard has failed, and P1 switches to the fallback lines	both

Opening cue: P2 shows Slide 1 and nods. P1 starts the timer and begins speaking.

Synchronised timeline

Elapsed	Presenter 1 says	Presenter 2 does	Audience sees	Transition / recovery cue
0:00–0:25	"Cities have far more streets than inspectors. Brooklyn has 23,448 street blocks, and in a typical week fewer than one in a hundred gets a pothole report. RoadWatch asks: can historical evidence show which blocks are most likely to get one next week, so a fixed inspection budget goes where it is needed?"	Nothing (Slide 1 is showing)	**Slide 1** · RoadWatch: 23,448 blocks · 0.9% of block-weeks · 71,609 reports	On "…where it is needed?": P2 presses →
0:25–0:55	"We used only real public data: over seventy-one thousand Brooklyn pothole reports from NYC 311 since 2020, the city's official street centreline for block shapes and road attributes, and NOAA weather. The hard part is location: thirty-nine percent of reports name an intersection or a vague stretch of street, so they could belong to several blocks, typically four."	—	**Slide 2** · three sources; matching bar 59.2% matched / 39.3% ambiguous	On "…typically four.": P2 presses →
0:55–1:27	"Every download is archived unchanged with checksums and validated, and an offline rebuild reproduces every table byte for byte. Reports are matched to blocks by street names and coordinates; ambiguous ones keep all their candidates. Weeks with too many unlocatable reports are marked unknown, never zero. And every feature is point-in-time: only 311 data published two days before the forecast, and weather from five days before."	—	**Slide 3** · four processing steps; t − 14 / t − 5 / t − 2 / t timeline	On "…five days before.": P2 presses →
1:27–1:57	"The model is LightGBM: gradient-boosted trees over 38 features, including recent report history, neighbouring blocks and road attributes like length and lanes. To test it honestly, we replayed history: sixteen monthly retrains from July 2025, each trained only on the past. The main comparison is a historical-rate baseline, which ranks blocks by how often they have been reported before."	—	**Slide 4** · walk-forward chart (16 retrains)	On "…reported before.": P2 presses →

Elapsed	Presenter 1 says	Presenter 2 does	Audience sees	Transition / recovery cue
1:57–2:27	"Across 67 held-out weeks with a 235-block list, LightGBM caught 13.4 percent of next week's reports, against 8.4 for the historical rate and 1 percent for a random list. Precision rose from 6.6 to 9.5 percent: a gain of 2.3 to 3.4 points, with 95 percent confidence. That is the strict target, where each report counts on exactly one block."	—	**Slide 5** · capture 13.4% vs 8.4% vs 1.0%; precision 9.5% vs 6.6% vs 0.9%	On "…exactly one block.": P2 presses →
2:27–2:40	"Turning ranks into decisions is deliberately simple: choose a weekly capacity, here 235 blocks, and take the top-ranked ones. Let's see it in the dashboard."	—	**Slide 6** · worklist screenshot + coverage chart	On "…in the dashboard." → **Alt+Tab** to the browser. *If the dashboard has failed: P2 stays on Slide 6 and raises a hand → fallback lines below*
2:40–2:52	"This is the week of 16 March 2026, a retrospective example, not a live recommendation. The blue lines are the 235 blocks to inspect, ranked in the table."	Hover over the worklist's first rows	**Dashboard** · map (blue = selected) · table rank 1 **17 AVE 11204 0.266**, rank 2 **FLATLANDS AVE 11236 0.250**	Map tiles missing → carry on (block lines still draw)
2:52–3:00	"If the budget doubles to 2 percent, the list grows to 469 blocks."	Sidebar: click **2% (469 blocks)**; after the metric updates, click **1% (235 blocks)**	Blocks to inspect **469** (P1/P2/P3 **94 / 141 / 234**), then back to **235**	Slow update → skip the switch back until the next step
3:00–3:15	"Every block shows its reasons: 17th Avenue had 59 reports naming it or its end intersections in the past year, 18 in the last month. That is evidence, not proof of a defect."	Scroll to **Why each block is listed** (block **#1 17 AVE**)	"#1 17 AVE (ZIP 11204, block 52878) – P1"; "59 report(s) … (18 in the last 30 days) …"; model drivers	Wrong block → choose **#1 17 AVE** in the *Block* dropdown
3:15–3:35	"One subtlety: what counts as catching a report depends on what an inspection covers. If inspecting a block also covers its end intersections, spreading inspections across intersections looks best. Without that assumption the strict list wins, so it is our default."	Scroll up; click view **Decision-policy comparison**; show the audit box and the table; point at **17.3%** (non-adjacent, intersection-node), then **13.5%** (strict top-K, fractional)	Four-definition table, Test 1%: strict top-K **13.5%** fractional / **16.4%** intersection-node; non-adjacent lists **12.3% / 11.3%** fractional, **17.3%** intersection-node	View slow → leave it; the words still stand
3:35–3:43	*(optional; skip if behind)* "That alternative was also proposed after we had seen test results, so we do not treat it as confirmed."	Keep the table on screen	Same table	On "…confirmed." (or "…our default.") → **Alt+Tab** to the slideshow, →

Elapsed	Presenter 1 says	Presenter 2 does	Audience sees	Transition / recovery cue
3:43–4:08	"Now the limits. In March 2026 reports roughly quadrupled. The ranking held up, but our weekly volume forecast fell 42 percent short, so capacity stays a human decision. We predict reports, not confirmed defects, and reports reflect who complains as well as road condition. We claim no savings and no failures prevented."	—	**Slide 7** · March 2026 chart; 843 reports/week; forecast 42% short; MAE 64.2 vs 64.1	On "…no failures prevented.": P2 presses →
4:08–4:30	"To make this deployable, we need inspection outcomes to learn real defects, better intersection locations, and a prospective trial that freezes the model and scores future weeks. RoadWatch shows that public data can rank where to look first: honestly measured, and ready for a field test. Thank you."	—	**Slide 8** · deployment table (data → implementation → metric)	**Ending cue:** "Thank you." P1 stops the timer. Slide 8 stays up for questions
4:30–5:00	*(contingency: silence or first question)*	—	Slide 8	Never start new material after 4:45

Fallback route if the live demo fails (from 2:40)

P2 stays on **Slide 6**, which shows a genuine screenshot of the same week and the coverage chart, and raises a hand. P1 replaces the demo lines with these (about 45 seconds):

> "Here is the same week in our dashboard, on the left: the blue lines are the 235 blocks to inspect, ranked in the table beside the map. Each listed block comes with its report history and the model's main drivers, and the list can be exported for inspectors. On the right is one honest subtlety: what counts as catching a report depends on what an inspection covers. If inspecting a block also covers its end intersections, the non-adjacent lists reach 17.3 percent; without that assumption the strict list is best, at 13.5 percent, so it is our default."

Then P2 advances to **Slide 7** and the talk continues from the 3:43 row. The timing gains about 15 seconds.

Partial failures (no need to switch):

- **Basemap tiles missing:** block lines, table and charts still work.
- **One slow view:** skip that step.
- **PowerPoint crash:** use the PDF tab; slide numbers are the same as page numbers.

Figures used on stage (all verified against saved files)

Figure	Source
23,448 blocks; 71,609 reports; 39.3% ambiguous (median 4 candidates)	`outputs/stage3/matching_diagnostics.json`
0.9% of block-weeks; 13.4% / 8.4% / 1.0% capture; 9.5% / 6.6% precision; 67 weeks	`outputs/stage4/test_results.csv`
+2.3 to +3.4 percentage points (95% CI)	`outputs/stage4/bootstrap_comparisons.json`
13.5%, 16.4%, 12.3%, 11.3%, 17.3% (Test, 1%)	`outputs/stage5/capture_definitions.csv` (shown verbatim by the dashboard)
469 blocks; 94 / 141 / 234; 17 AVE (59 reports, 18 in 30 days)	dashboard, verified in `outputs/final/verification/run_of_show_values.json`
843 per week; 42% short; MAE 64.2 vs 64.1	`outputs/stage5/volume_forecast_results.json`

Timing evidence

- **Speech length:** 555 words (536 without the optional line), which is 4:16 at 130 words per minute.
- **Synthetic-voice read-through:** 5:19 at a slow 104 words per minute (`outputs/final/verification/speech_timing.json`). That is a worst case; rehearse once with a human voice.
- **Checkpoints:** reach Slide 6 by about 2:35 and Slide 7 by 3:50.
- **If you are behind:** apply the trims listed in the speech document.